Worker Rights in Collective Bargaining

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Motivation & Research Question

- Collective bargaining agreements (CBAs) run to dozens of pages of legally binding plain-text clauses, yet economists have studied mainly wages and enumerated benefits.
- Is the rest of the text merely 'cheap talk', or do the legal rights it grants have real economic value to workers and firms?
  - Property-rights theory of the firm (Grossman-Hart; Aghion-Tirole): rights clauses reallocate residual decision authority within the firm, not just compensation.
- Goal: measure the full legal content of CBAs from text, then study empirically what determines worker rights and what they are worth.
- Two questions: (1) How can worker rights be extracted and validated from contract language? (2) How do rights respond to taxes and labor-market conditions, and what is their monetary value?

Data: A New Corpus of 30,000 Canadian CBAs

- 32,402 English-language CBAs from Employment and Social Development Canada, 1986-2015 (French/Quebec contracts excluded).
- 7,572 companies (~4 contracts each), 10 provinces + 3 territories, 11 sectors, 606 NAICS industries; ~half private sector.
  - Why Canada: common-law, decentralized FIRM-level bargaining, comprehensive contract database, and province-level variation in labor regulation.
- Metadata per contract: company, union, location, sector, employees, signing/effective/expiry dates.
- Matched to province-year implicit income tax rates, Labour Force Survey employment/wages (4M+ workers), and the World Management Survey.

NLP Method I: Structural Parsing of Rights & Duties

Table 2: Summary Statistics: What do Union Contracts Consist of?

Agent	Obligations		Rights		Permissions		Prohibitions		Total	
	Count	%	Count	%	Count	%	Count	%	Count	%
Worker	666021	20.9	730327	22.9	266278	8.4	99795	3.1	1,762,421	**55.3**
Firm	788857	24.7	27382	0.9	108114	3.4	49099	1.5	973,452	**30.5**
Union	224119	7.0	68410	2.1	62912	2.0	17786	0.6	373,227	**11.7**
Manager	55518	1.7	7512	0.2	12537	0.4	3107	0.1	78,674	**2.5**
Total	1,734,515	**54.4**	833,631	**26.2**	449,841	**14.1**	169,787	**5.3**	3,187,774	100.0

Note: Table 2: clause counts and shares by agent (rows) x statement type (columns). Worker rights are 22.9% of all clauses; firm rights just 0.9%.

- Structural, not bag-of-words: small wording changes flip legal meaning ('may' vs 'must', 'shall' vs 'shall not').
- spaCy dependency parse of 10.8M sentences into Subject-Verb-Object clauses.
  - Subject -> agent: worker, firm, union, manager.
  - Deontic modal verb -> right, permission, obligation, or prohibition (Hohfeld 1913).
- 'Employees shall receive...' = worker right; 'the employer shall provide...' = firm obligation.
- Contracts mainly do three things: firm obligations, worker rights, worker obligations.

NLP Method II: From Clauses to Worker-Rights Topics

- Each worker-rights clause is embedded with a transformer sentence encoder (768-dim, context-sensitive), then grouped by k-means into 30 topics.
- Advantages over LDA: synonyms share meaning, context-sensitive, one clause -> one topic (better for short sentences).
- Topics span both wage-like benefits (pay, vacations, sick leave) and control rights (scheduling, seniority, grievance); scheduling is the largest.
  - 'Sim-to-Wages' score: embedding proximity of each topic to simulated priced-benefit clauses (from the BLS Compensation Survey codebook) -> ranks priced vs. control rights.
- GPT-4 generates a plain-language summary of each topic; 30 narrow topics hand-aggregated into 7 broad categories.
- Discussion (Sec 4.6): parser preferred over dictionaries, supervised ML, and LLMs -- transparent, reproducible, and embeds legal knowledge in its rules.

Validation: An LLM Confirms Worker Rights Are Pro-Worker

Table 4: LLM Validation of Worker Rights as Favoring Workers

Clause Type	Clause Frequency (%)	Pro-Worker Frequency (%)
Worker Right	**22.9**	**79.7**
Worker Permission	8.4	67.3
Union Right	2.1	65.7
Firm Obligation	24.7	60.7
Manager Right	0.2	58.4
Worker Obligation	20.9	53.1
Manager Obligation	1.7	44.5
Union Permission	2.0	43.2
Worker Prohibition	3.1	41.6
Manager Prohibition	0.1	38.8
Firm Prohibition	1.5	36.5
Firm Permission	3.4	35.5
Manager Permission	0.4	34.3
Union Obligation	7.0	31.3
Union Prohibition	0.6	28.3
Firm Right	0.9	28.2

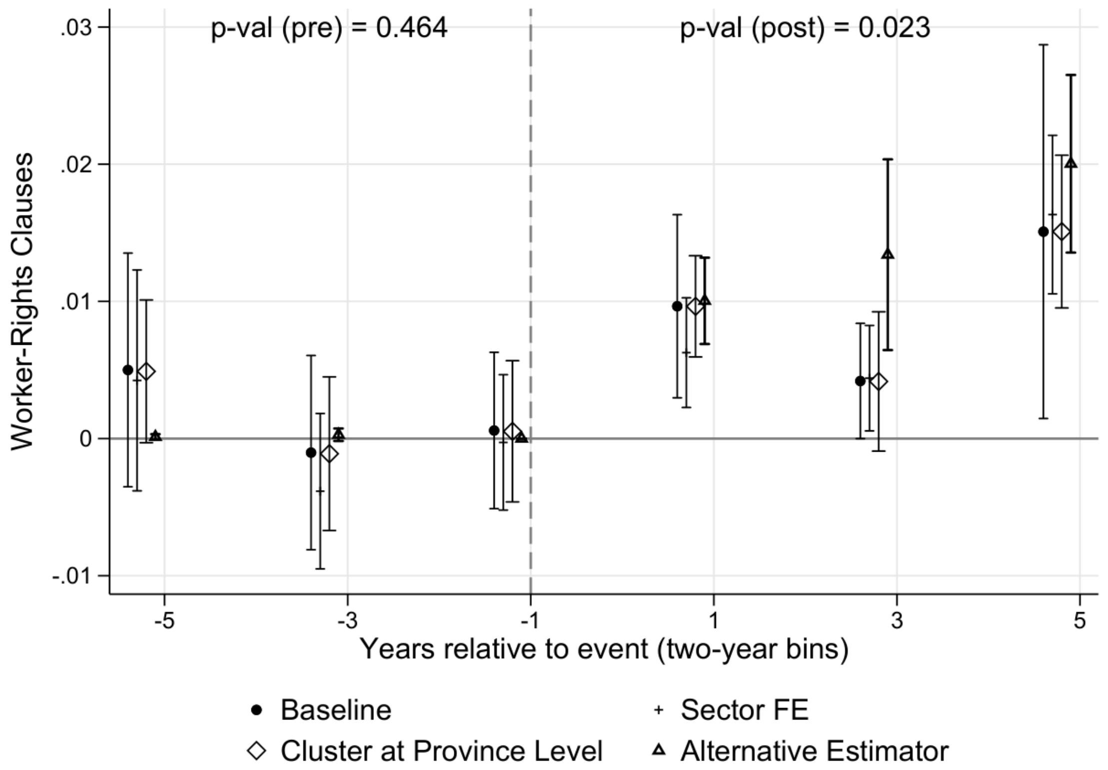
Note: Table 4: GPT-4o pairwise comparisons of 24,000 clause pairs ('which is more of an entitlement/benefit for workers?'). Clauses parsed as worker rights are judged pro-worker 79.7% of the time -- far above any other type; firm rights rank last (28.2%). LLM matches human labels 83% of the time when one clause is a worker right. Rights also correlate with pro-worker HR practices in the World Management Survey (Figure 2).

Empirical Strategy: Determinants of Worker Rights

- Outcome: share of worker-rights clauses in a contract (province p, sector s, firm i, year t).
- Treatment 1 -- log province-year implicit labor income tax rate: alters the relative price of taxed wages vs. untaxed rights.
- Treatment 2 -- leave-one-out sector employment rate: a labor-demand shock proxying workers' outside options / bargaining power.
  - Identification: province-by-sector and year-by-sector fixed effects; fixed contract-expiration schedule limits endogenous timing; controls for other labor laws, NDP government.
- Callaway-Sant'Anna staggered event studies test parallel trends; IV isolates province tax changes driven by FEDERAL rate changes (Gruber-Saez; F=65).
- Standard errors clustered by province-sector.

Result 1: Higher Income Taxes Raise Worker Rights

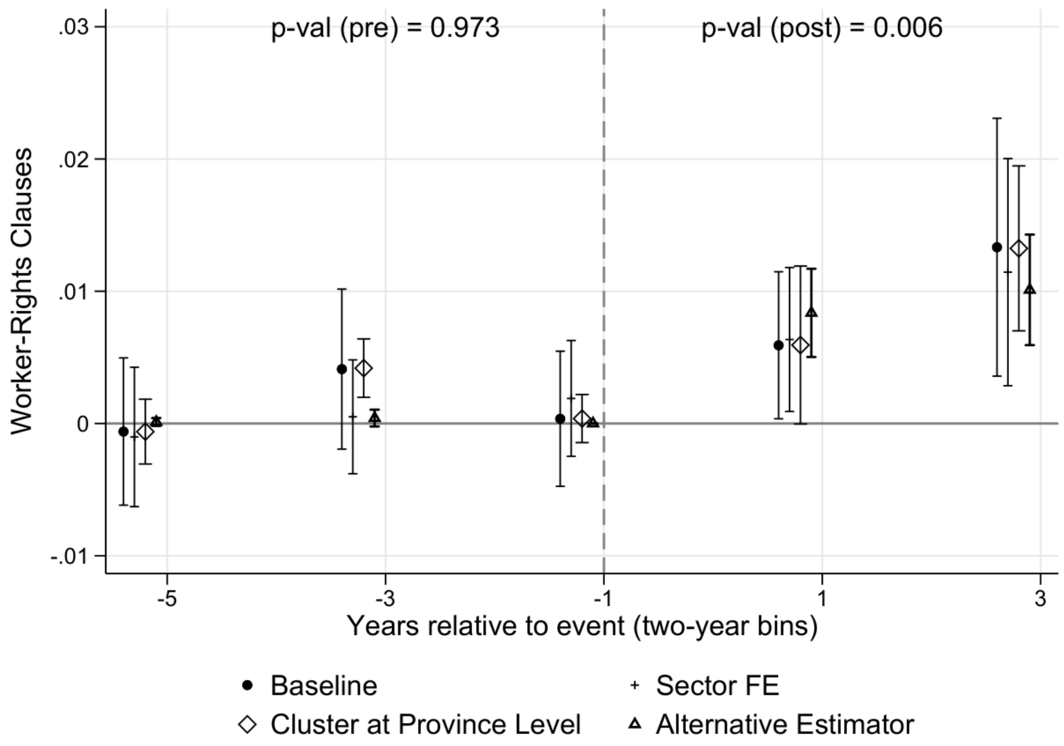
Figure 4: Event Study: Income Tax Increases and Worker Rights



Note: Figure 4: Event study (Callaway-Sant'Anna) of large province labor-tax increases on the worker-rights share. Flat pre-trends (pre p=0.464), significant rise after the event (post p=0.023), robust across estimators. Consistent with substitution from taxed wages toward untaxed rights (cf. Gruber-Poterba 1994). In OLS, a 10% tax rise raises the rights share by ~0.23 s.d., with no effect on non-rights clauses.

Result 2: Stronger Outside Options Raise Worker Rights

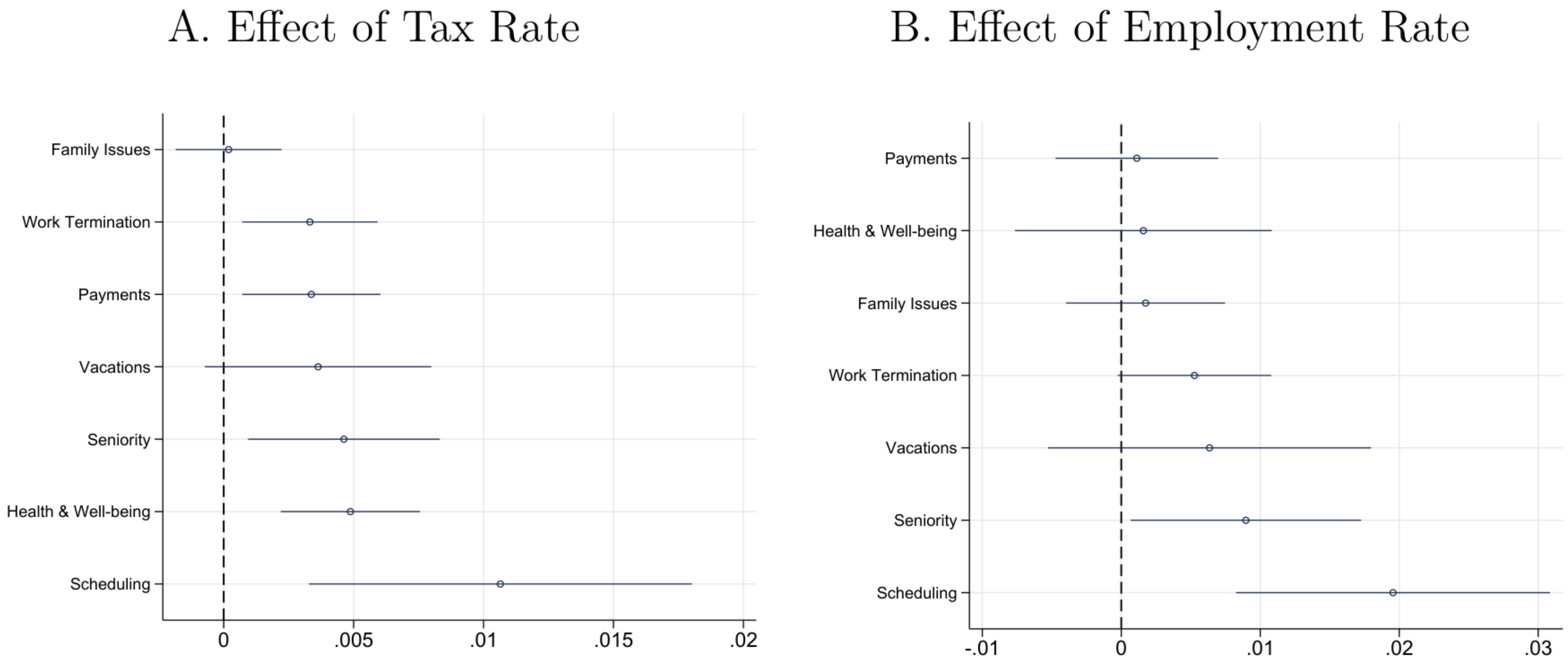
Figure 5: Event Study: Employment Increases and Worker Rights



Note: Figure 5: Event study of large increases in the leave-one-out sector employment rate (a positive shock to workers' outside options). Flat pre-trends (pre p=0.973) and a significant post-event increase in worker rights (post p=0.006). Consistent with a shift in relative bargaining power toward workers; mirrors Lagos (2025) for Brazil. Non-worker-rights clauses are unaffected.

Which Rights? Scheduling and Seniority Respond Most

Figure 6: Effect of Tax Rate and Employment Rate on Worker-Rights Clauses, by Category



Note: Figure 6: Effects of taxes (Panel A) and employment (Panel B) on worker-rights shares by broad category. Largest effects are on Scheduling for both, plus Seniority and Health & Well-being. Effects on unpriced CONTROL rights (scheduling, seniority) are at least as strong as on cash-like benefits -- yet these are invisible in standard datasets.

Valuation & Takeaways

- Combining responses: tax increases raise the rights share but lower union pre-tax wages relative to non-union -> unions substitute wages for rights.
- A one-standard-deviation increase in the worker-rights share is worth about 5.7% of wages -- in line with amenity valuations (Mas-Pallais 8%; Dube et al. 6%).
- Worker rights behave like non-wage compensation, but are legally durable: backed by grievance, arbitration, and strike rights.
- Method contribution: a scalable, transparent, reproducible grammar-based pipeline to measure legal content of contracts -- extendable beyond CBAs.
- Implication: analyzing contract TEXT is essential to capture control rights (scheduling, seniority) that matter to workers but are missed by wage/benefit data.